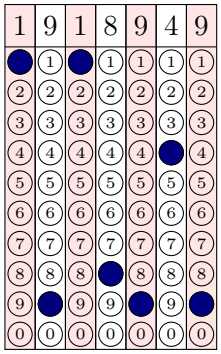
TEST VARIANTI RAQAMI: **1918949**

MATEMATIKA

Ushbu test varianti 30 ta test topshirig‘idan iborat. Uni bajarishga 1 soat vaqt belgilash tavsiya etiladi. Natijani bilish uchun javoblaringizni javoblar varaqasiga belgilab, rasmini **@dtmxizmatbot** orqali jo‘nating. Sizga omad tilaymiz!

TEST VARIANTI RAQAMINI JAVOBLAR VARAQASIGA TO‘G‘RI KO‘CHIRING!



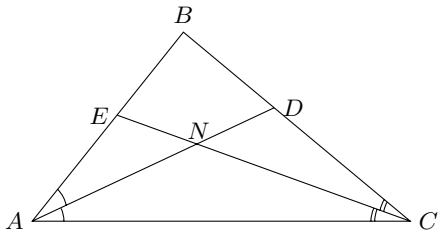
1. $x = 3,61(91)$, $y = 3,62$, $z = 3,6(191)$ va $t = 3,619(1)$ sonlarini kamaytirish tartibida yozing.

A) $x > z > y > t$ B) $y > t > z > x$
C) $y > x > t > z$ D) $y > x > z > t$

2. $x^3 + mx^2 - 4x + n = 0$ tenglamaning ildizlari $x_1 = 3$ va $x_2 = -2$ bo‘lsa, $3m + n$ ning qiymatini toping.

A) *3* B) *6* C) *−12* D) *8*

3. Rasmda ABC uchburchak, uning AD va CE bissektrisalari tasvirlangan. Agar $\angle ABC = 108°$ bo‘lsa, $\angle ENA$ burchakni toping.



A) 28° B) 36° C) 34° D) 38°

4. Hisoblang: $\dfrac{512 \cdot \left(2^4\right)^3}{\left(2^7\right)^2 \cdot 128} \cdot \left(2^{-4}\right)^{-3} \cdot 4^{-5}$

A) $\dfrac{1}{2}$ B) *1* C) *8* D) *4*

5. Qutida 4 ta qora va 5 oq shar bor. Qutidan tavakkaliga olingan ikkita sharning ikkalasi ham oq shar bo‘lishi ehtimolligini toping.

A) $\dfrac{1}{3}$ B) $\dfrac{5}{18}$ C) $\dfrac{2}{9}$ D) $\dfrac{1}{6}$

6. Hisoblang: $\cos^6 11° - 3\cos^4 11° + 2\cos^2 11° + \sin^6 11° - \sin^2 11°$

A) 1 B) 2 C) −1 D) 0

7. $\log_2^2 x + 2 \geq 3\log_2 x$ tengsizlikni yeching.

A) $[2; 4]$ B) $\left(0; \dfrac{1}{4}\right] \cup \left[\dfrac{1}{2}; \infty\right)$
C) $(0; 2] \cup [4; \infty)$ D) $(-\infty; 2] \cup [4; \infty)$

8. $f(x) = (k+2)x + 2$ funksiya k ning qanday qiymatlarida o‘suvchi bo‘ladi?

A) $k < 0$ B) $k \in R$ C) $k < -2$
D) $k > -2$

9. Ifodani soddalashtiring:

$$\frac{a^{-\frac{7}{2}} - \dfrac{1}{a^2}}{a^{-\frac{5}{2}} - a^{-1}} + a^2 - \frac{1}{a} - 1.$$

A) $a - 1$ B) $1 - a^2$ C) $a + 1$ D) $a^2 - 1$

10. $f(x) = ax^2 + bx + c$ funksiyaning grafigi $A(-2; 5)$, $B(3; 10)$ va $C(-1; -2)$ nuqtalardan o‘tadi. $f'(3) - 2f(2)$ ning qiymatini toping.

A) *4* B) *−3* C) *6* D) *9*

11. $\dfrac{\left(\sqrt{3x-7}\right)^2 - 2}{x-3} \leq \dfrac{3 - \left(\sqrt{x}\right)^2}{x-3}$ tengsizlikni yeching.

A) *yechimga ega emas* B) $x \in R$
C) $\left[0; 2\dfrac{1}{3}\right]$ D) $\left[2\dfrac{1}{3}; 3\right) \cup (3; \infty)$

12. n ning qanday qiymatida $2n + 3$ ning 60%i 45 ga teng bo‘ladi?

A) *36* B) *42* C) *24* D) *32*

13. Qirralari 2 dm, 3 dm va 4 dm bo‘lgan to‘g‘ri burchakli parallelepiped shaklidagi quti ichiga eng ko‘pi bilan qirrasi 7 cm bo‘lgan kublardan nechtasini joylashtirish mumkin?

A) *45* B) *42* C) *69* D) *40*

© Davlat test markazi, 2019

14. $(x;y)$ sonlar juftligi $\begin{cases} \dfrac{3}{x+2}+\dfrac{7}{6+y}=1\dfrac{1}{4} \\ \dfrac{10}{x+2}-\dfrac{14}{y+6}=-\dfrac{1}{2} \end{cases}$ tenglamalar sistemasini qanoatlantiradi. $(x-y)^y$ ning qiymatini toping.

A) *36* B) *16* C) *125* D) *1*

15. Integralni hisoblang: $\int (7x-3)\cos 4x dx$.

A) $\dfrac{(7x-3)\sin 4x}{4}+\dfrac{7}{16}\cos 4x+C$

B) $\dfrac{(7x-3)\cos 4x}{4}+\dfrac{7}{16}\sin 4x+C$

C) $\dfrac{(7x-3)\sin 4x}{8}+\dfrac{7}{4}\cos 4x+C$

D) $\dfrac{(7x-3)\sin 4x}{4}-\dfrac{7}{16}\cos 4x+C$

16. Asosi $4\sqrt{2}$ ga teng va unga yopishgan burchaklari 30° va 45° bo‘lgan uchburchakning yuzasini toping.

A) $4\sqrt{\sqrt{3}+2}$ B) $16\left(\sqrt{3}-1\right)$
C) $8\left(\sqrt{3}-1\right)$ D) $8\sqrt{\sqrt{3}+1}$

17. $1\dfrac{1}{6};1\dfrac{1}{2};1\dfrac{5}{6};...$ sonlar arifmetik progressiyaning hadlari bo‘lsa, uchinchi hadi o‘ninchi hadining necha foizini tashkil etadi?

A) *24* B) *44* C) *66* D) *52*

18. To‘g‘ri burchakli uchburchakka ichki va tashqi aylanalar chizilgan. Agar uchburchakning katetlari 6 va 8 ga teng bo‘lsa, aylanalar markazlari orasidagi masofani toping.

A) $\sqrt{5}$ B) 2 C) $\sqrt{3}$ D) $\sqrt{6}$

19. Agar $\vec{a}$, $\vec{b}$ vektorlar uchun $2\vec{a}+\vec{b}=6\vec{i}+9\vec{j}$ va $\vec{a}+2\vec{b}=-3\vec{i}+6\vec{j}$ o‘rinli bo‘lsa, $\vec{a}$ vektorni toping.

A) $\bar{a}\,(5;-4)$ B) $\bar{a}\,(-5;-4)$ C) $\bar{a}\,(-5;4)$
D) $\bar{a}\,(5;4)$

20. $f(x)=e^{-3x+2}\cdot\sin 3x$ funksiyaning hosilasini toping.

A) $3e^{-3x+2}\cdot(\cos 3x+\sin 3x)$

B) $-3e^{-3x+2}\cdot(\cos 3x-\sin 3x)$

C) $3e^{-3x+2}\cdot(\cos 3x-\sin 3x)$

D) $-3e^{-3x+2}\cdot(\cos 3x+\sin 3x)$

21. Farxod, Anvar va Jahongir birgalikda 57,8 kg uzum yig‘ishdi. Anvar Farxodga nisbatan 2,4 kg ko‘p, Jahongir esa Farxod bilan Anvarning birgalikdagi yig‘ganidan 20,6 kg kam yig‘gan. Anvar va Jahongir birgalikda necha kg uzum yig‘ishgan?

A) *39,4* B) *36,8* C) *40,2* D) *38,8*

22. Agar $2m+3n=72$ $(m,n\in N)$ bo‘lsa, n ning eng katta qiymatini toping.

A) *22* B) *20* C) *18* D) *24*

23. Ko‘paytirishni bajaring:
$\left(3x^2-\dfrac{1}{2}y\right)\cdot\left(3x^2+\dfrac{1}{2}y\right)$

A) $9x^4-\dfrac{1}{8}y^2$ B) $18x^4-\dfrac{1}{4}y^2$ C) $9x^4-\dfrac{1}{4}y^2$
D) $9x^4-\dfrac{1}{2}y^2$

24. $8^{5-\frac{x^2}{3}}<\dfrac{1}{2}$ tengsizlikning eng kichik natural yechimini toping.

A) *4* B) *5* C) *3* D) *6*

25. $a=0,6^{\frac{1}{3}}\cdot 1,3^{-\frac{2}{5}}$, $b=0,7^{-\frac{2}{3}}\cdot 0,3^{-\frac{1}{5}}$ va $c=1,8^{\frac{1}{3}}\cdot 0,3^{-\frac{2}{5}}$ sonlardan qaysilari 1 dan katta?

A) *a va c* B) *faqat b* C) *b va c* D) *a va b*

26. Beshburchakli prizmada nechta turli diagonal kesim mavjud?

A) *2* B) *10* C) *5* D) *3*

27. $\dfrac{a^3b+2a^2b-3ab}{a^3+5a^2+6a}\cdot\left(\dfrac{1-a^2}{a^2+3a+2}\right)^{-1}+2b$ ifodaning $a=\dfrac{1}{3}$, $b=-6$ dagi qiymatini toping.

A) $\dfrac{1}{3}$ B) *9* C) $-6\dfrac{1}{3}$ D) -6

28. 30 ta o‘quvchidan $3x+1$ tasi rus tilini, $2x-1$ tasi ingliz tilini, $x-1$ tasi rus tilini ham ingliz tilini ham biladi. 5 tasi esa rus tilini ham ingliz tilini ham bilmaydi. Nechta o‘quvchi rus tilini biladi?

A) *17* B) *19* C) *11* D) *13*

29. Agar $\sin 5^\circ\cdot\cos 5^\circ\cdot\cos 10^\circ\cdot\cos 20^\circ\cdot\cos 40^\circ=m$ bo‘lsa, $\dfrac{1}{2}\sin 80^\circ$ ni m orqali ifodalang.

A) $-4m$ B) $4m$ C) $8m$ D) $16m$

30. $y=2x^2-8x+11$ parabola uchining koordinatalari yig‘indisini toping.

A) *6* B) *5* C) *8* D) *7*

© Davlat test markazi, 2019

Test varianti “Test topshiriqlari to‘plami 2019” asosida shakllantirilgan:

1 – 10-bet 8	7 – 57-bet 22	13 – 110-bet 34	19 – 115-bet 7	25 – 15-bet 61
2 – 63-bet 37	8 – 71-bet 13	14 – 64-bet 46	20 – 79-bet 25	26 – 106-bet 1
3 – 92-bet 17	9 – 27-bet 53	15 – 89-bet 38	21 – 22-bet 56	27 – 51-bet 60
4 – 25-bet 32	10 – 84-bet 64	16 – 99-bet 70	22 – 7-bet 3	28 – 122-bet 26
5 – 125-bet 24	11 – 66-bet 3	17 – 31-bet 15	23 – 47-bet 21	29 – 39-bet 43
6 – 41-bet 67	12 – 17-bet 18	18 – 104-bet 101	24 – 53-bet 17	30 – 75-bet 52

© Davlat test markazi, 2019